20. Function Machines

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — One-step machine:

A function machine applies the rule ×6 to its input. What is the output when the input is 8?

Answer: 8 × 6 = **48**.

Example 2 — Two-step (chained) machine:

A function machine: input → +4 → ×3 → output. What is the output when the input is 6?

Answer: 6 + 4 = 10; 10 × 3 = **30**.

Example 3 — Working backwards:

A function machine: input → ×2 → −5 → output. The output is 13. Find the input.

Answer: Reverse the steps: 13 + 5 = 18; 18 ÷ 2 = **9**.

Example 4 — Finding an unknown rule from two pairs:

A function machine turns 4 into 17 and 7 into 26 using the same two-step rule "×a, then +b". Find the rule (give both a and b).

Answer: Going from 4 to 7 is +3, and the output rises from 17 to 26, which is +9. So a = 9 ÷ 3 = **3**. Then 3 × 4 + b = 17, so b = 17 − 12 = **5**. The rule is ×3, then +5.

Example 5 — Multi-step word problem:

Hiring a hall costs £5, plus £3 for every guest. What is the total hire cost for 8 guests?

Answer: Treat this as a function machine: input → ×3 → +5 → output. 8 × 3 = 24; 24 + 5 = **£29**.

Foundation

15 questions

1. A function machine applies the rule ×7 to its input. What is the output when the input is 8?

2. A function machine applies the rule +12 to its input. What is the output when the input is 9?

3. A function machine applies the rule −9 to its input. What is the output when the input is 20?

4. A function machine applies the rule ÷6 to its input. What is the output when the input is 42? A) 5 B) 6 C) 7 D) 8 E) 9

5. A two-step function machine: input → +4 → ×2 → output. What is the output when the input is 6?

6. A two-step function machine: input → ×3 → −2 → output. What is the output when the input is 7? A) 17 B) 18 C) 19 D) 20 E) 21

7. A function machine applies the rule ×5 to its input. If the output is 35, what was the input?

8. A function machine applies the rule +18 to its input. If the output is 25, what was the input? A) 5 B) 6 C) 7 D) 8 E) 9

9. A function machine applies the rule −7 to its input. If the output is 9, what was the input?

10. A function machine applies the rule ÷3 to its input. If the output is 8, what was the input?

11. Select ALL that apply: a function machine applies the rule ×4 to each input. Which of these input→output pairs are correct? A) 3→12 B) 4→16 C) 5→19 D) 6→24 E) 7→29

12. A two-step function machine: input → ×5 → −4 → output. What is the output when the input is 4?

13. A two-step function machine: input → +9 → ÷2 → output. What is the output when the input is 7?

14. A function machine takes an input, triples it, then adds 7. What is the output when the input is 8? A) 24 B) 27 C) 29 D) 31 E) 33

15. A function machine applies the rule −15 to its input. What is the output when the input is 15?

Standard

20 questions

1. A two-step function machine: input → ×2 → +7 → output. What is the output when the input is 9?

2. A two-step function machine: input → −6 → ×3 → output. What is the output when the input is 14?

3. A two-step function machine: input → ÷4 → −3 → output. What is the output when the input is 20?

4. A machine takes an input, adds 5, then multiplies by 4. If the output is 36, what was the input? A) 2 B) 3 C) 4 D) 5 E) 6

5. A function machine: input → ×3 → −11 → output. The output is 16. Find the input.

6. A function machine turns 3 into 13 and 6 into 22 using the same two-step rule "×a, then +b". Find the rule (give both a and b).

7. A function machine: input → ×8 → output. What is the output when the input is 0.25?

8. Jamal puts a number into a machine. It doubles the number, then adds 9. The output is 27. What number did Jamal put in? A) 7 B) 8 C) 9 D) 10 E) 11

9. A function machine: input → +11 → output. What is the output when the input is −6?

10. Select ALL that apply: a function machine applies the rule "subtract 5, then double" to inputs. Which input→output pairs are correct? A) 7→4 B) 9→11 C) 11→12 D) 13→17 E) 15→20

11. A function machine: input → ×3 → output. What is the output when the input is 2/7?

12. A machine triples a number and then subtracts 7. If the result is 11, what was the original number?

13. A function machine: input → ×0.5 → output. What is the output when the input is 17? A) 7.5 B) 8 C) 8.5 D) 9 E) 9.5

14. A function machine: input → ÷5 → +8 → output. What is the output when the input is −15?

15. A function machine takes a number, adds 1/3 to it, then doubles the result. What is the output when the input is 2/3?

16. A function machine applies the rule ×4 to its input. What is the output when the input is 12?

17. A two-step function machine: input → ×6 → −2 → output. What is the output when the input is 5?

18. A function machine applies the rule ÷8 to its input. What is the output when the input is 56?

19. A two-step function machine: input → +13 → ÷2 → output. What is the output when the input is 9?

20. A function machine applies the rule −20 to its input. What is the output when the input is 50?

Hard

5 questions

1. A taxi charges £4 as a starting fee, plus £2.50 for every mile. What is the total cost for a 9-mile journey?

2. A function machine applies the rule "×4, then +b". An input of 5 gives an output of 23. What is b?

3. A function machine turns 2 into 11 and 5 into 20 using the same two-step rule "×a, then +b". Find the rule (give both a and b).

4. A number is put into a machine. It adds 9 to the number, then divides the result by 3. The final result is 7. What number was put in?

5. Select ALL that apply: a function machine applies the rule “halve, then add 7” to inputs. Which input→output pairs are correct? A) 10→12 B) 12→16 C) 14→14 D) 16→14 E) 18→16

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